

OCRStudioSDK Library Reference

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1 Class Documentation

1.1 ocrstudio::OCRStudioSDKDelegate Class Reference

Public Member Functions

- virtual [~OCRStudioSDKDelegate](#) ()=default
Virtual destructor.
- virtual void [Callback](#) (const char *json_message)=0
Callback for receiving messages from processing session.

1.1.1 Detailed Description

Definition at line 19 of file [ocr_studio_delegate.h](#).

1.1.2 Member Function Documentation

1.1.2.1 [Callback\(\)](#) virtual void ocrstudio::OCRStudioSDKDelegate::Callback (
const char * json_message) [pure virtual]

Callback for receiving messages from processing session.

Parameters

<i>json_message</i>	- callback message encoded in JSON string
---------------------	---

1.2 ocrstudio::OCRStudioSDKException Class Reference

Public Member Functions

- virtual [~OCRStudioSDKException](#) ()
Non-trivial destructor.
- [OCRStudioSDKException](#) (const char *type, const char *msg)
Main constructor.
- [OCRStudioSDKException](#) (const [OCRStudioSDKException](#) ©)
Copy constructor.
- const char * [Type](#) () const
Returns exception type.
- const char * [Message](#) () const
Returns exception message.

Private Attributes

- char * [type_](#)
stored exception type
- char * [msg_](#)
stored exception message

1.2.1 Detailed Description

Definition at line 19 of file [ocr_studio_exception.h](#).

1.3 ocrstudio::OCRStudioSDKImage Class Reference

Bitmap image class.

Public Member Functions

- virtual [~OCRStudioSDKImage](#) ()=default
Default destructor.
- virtual [OCRStudioSDKImage](#) * [DeepCopy](#) () const =0
Copies an image with copying of all pixels.
- virtual [OCRStudioSDKImage](#) * [ShallowCopy](#) () const =0
Copies an image without copying the pixels, retaining internal memory reference. The operations with the copied image will be invalid after the source is deleted.
- virtual void [Clear](#) ()=0
Clears the internal structure of the image.
- virtual int [ExportPixelBufferLength](#) () const =0
Returns the required size of the export pixel buffer.
- virtual int [ExportPixelBuffer](#) (unsigned char *export_buffer, int export_buffer_length) const =0
Copies the pixels into an external buffer. For any image the exported buffer pixels will have 8-bit channels (0 means lowest intensity, 255 means highest intensity). 1-channel images are exported as grayscale, 3-channel images are exported as RGB, other images are copied as-is.
- virtual [OCRStudioSDKString](#) [ExportBase64JPEG](#) () const =0

- Exports image as a JPEG buffer encoded in base64.*

 - virtual void [Scale](#) (int width, int height)=0

Scales the internal image to a new size.

 - virtual [OCRStudioSDKImage](#) * [DeepCopyScaled](#) (int width, int height) const =0
- Copies the image with scaling to a new size.*
- virtual void [CropByQuad](#) (const char *quad_json, int width, int height)=0
- Crops an image quadrilateral to a new image, with a new provided size. If width or height is less or equal to zero, the size will be calculated approximately based on an input quadrilateral.*
- virtual [OCRStudioSDKImage](#) * [DeepCopyCroppedByQuad](#) (const char *quad_json, int width, int height) const =0
- Copies an image cropped by a quadrilateral, with a new provided size. If width or height is less or equal to zero, the size will be calculated approximately based on an input quadrilateral.*
- virtual void [CropByRect](#) (int x, int y, int width, int height)=0
- Crops an image to a rectangular region.*
- virtual [OCRStudioSDKImage](#) * [DeepCopyCroppedByRect](#) (int x, int y, int width, int height) const =0
- Copies an image cropped to a rectangular region.*
- virtual [OCRStudioSDKImage](#) * [ShallowCopyCroppedByRect](#) (int x, int y, int width, int height) const =0
- Shallow-copies an image cropped to a rectangular region. Operations on the resulting image are invalid after the source image is deleted.*
- virtual void [RotateByNinety](#) (int num_rotations)=0
- Rotates the image clockwise by 90 degrees.*
- virtual [OCRStudioSDKImage](#) * [DeepCopyRotatedByNinety](#) (int num_rotations) const =0
- Copies the image rotated clockwise by 90 degrees.*
- virtual int [Width](#) () const =0
- Image width in pixels.*
- virtual int [Height](#) () const =0
- Image height in pixels.*
- virtual int [BytesPerLine](#) () const =0
- Size of the image row in bytes, including alignment.*
- virtual int [Channels](#) () const =0
- The number of channels per pixel.*
- virtual void * [UnsafeBufferPtr](#) () const =0
- Gets the pointer to the pixels buffer.*
- virtual bool [OwnsPixelData](#) () const =0
- Whether this instance owns and will release pixel data.*
- virtual void [ForcePixelDataOwnership](#) ()=0
- Forces pixel data ownership - for shallow images, copies all pixels.*

Static Public Member Functions

- static int [PagesCount](#) (const char *filename)

For multi-page images, returns the number of pages in an image file.

- static [OCRStudioSDKString](#) [PageName](#) (const char *filename, int page_number)

For multi-page images, returns the filename of a particular page.

- static [OCRStudioSDKImage](#) * [CreateEmpty](#) ()

Creates an empty image.

- static [OCRStudioSDKImage](#) * [CreateFromFile](#) (const char *filename, int page_number=0, int max_width=25000, int max_height=25000)

Creates an image from file.

- static [OCRStudioSDKImage](#) * [CreateFromFileBuffer](#) (unsigned char *data, int data_size, int page_number=0, int max_width=25000, int max_height=25000)

Creates an image from file loaded in a buffer.

- static `OCRStudioSDKImage * CreateFromBase64FileBuffer` (const char *base64_data, int page_number=0, int max_width=25000, int max_height=25000)

Creates an image from file loaded in a buffer encoded in base64.

- static `OCRStudioSDKImage * CreateFromPixelBuffer` (unsigned char *data, int data_size, int width, int height, int bytes_per_line, int bytes_per_channel, OCRStudioSDKPixelFormat pixel_format)

Creates an image from a pixel buffer, the content is copied.

- static `OCRStudioSDKImage * CreateFromBuffer` (unsigned char *data, int data_size, int width, int height, int bytes_per_line, int channels)

Creates an image from a buffer, the content is copied.

- static `OCRStudioSDKImage * CreateFromYUVSimple` (unsigned char *yuv_data, int yuv_data_size, int width, int height)

Creates an image from a simple YUV NV21 buffer.

- static `OCRStudioSDKImage * CreateFromYUV` (unsigned char *y_plane, int y_plane_size, int y_plane_row_stride, int y_plane_pixel_stride, unsigned char *u_plane, int u_plane_size, int u_plane_row_stride, int u_plane_pixel_stride, unsigned char *v_plane, int v_plane_size, int v_plane_row_stride, int v_plane_pixel_stride, int width, int height, OCRStudioSDKYUVFormat yuv_format)

Creates an image from a universal YUV buffer.

1.3.1 Detailed Description

Bitmap image class.

Definition at line 50 of file [ocr_studio_image.h](#).

1.3.2 Member Function Documentation

1.3.2.1 PagesCount() static int ocrstudio::OCRStudioSDKImage::PagesCount (const char * filename) [static]

For multi-page images, returns the number of pages in an image file.

Parameters

<i>filename</i>	- path to an image file
-----------------	-------------------------

Returns

The number of pages in an image

1.3.2.2 PageName() static `OCRStudioSDKString` ocrstudio::OCRStudioSDKImage::PageName (const char * filename, int page_number) [static]

For multi-page images, returns the filename of a particular page.

Parameters

<i>filename</i>	- Filename of a particular image page
<i>page_number</i>	- page number, starting with 0

Returns

The string representation of a page filename

1.3.2.3 CreateEmpty() static `OCRStudioSDKImage*` ocrstudio::OCRStudioSDKImage::CreateEmpty ()
[static]

Creates an empty image.

Returns

Pointer to a new image, the ownership is relinquished.

1.3.2.4 CreateFromFile() static `OCRStudioSDKImage*` ocrstudio::OCRStudioSDKImage::CreateFromFile (
const char * *filename*,
int *page_number* = 0,
int *max_width* = 25000,
int *max_height* = 25000) [static]

Creates an image from file.

Parameters

<i>filename</i>	- path to an image file (png, jpg, tif)
<i>page_number</i>	- page number, starting with 0
<i>max_width</i>	- maximum image width in pixels (0 for unrestricted)
<i>max_height</i>	- maximum image height in pixels (0 for unrestricted)

Returns

Pointer to a new image, the ownership is relinquished.

1.3.2.5 CreateFromFileBuffer() static `OCRStudioSDKImage*` ocrstudio::OCRStudioSDKImage::CreateFrom
FromFileBuffer (
unsigned char * *data*,
int *data_size*,

```

int page_number = 0,
int max_width = 25000,
int max_height = 25000 ) [static]

```

Creates an image from file loaded in a buffer.

Parameters

<i>data</i>	- pointer to a loaded file buffer
<i>data_size</i>	- size of the loaded file buffer
<i>page_number</i>	- page number, starting with 0
<i>max_width</i>	- maximum image width in pixels (0 for unrestricted)
<i>max_height</i>	- maximum image height in pixels (0 for unrestricted)

Returns

Pointer to a new image, the ownership is relinquished.

1.3.2.6 CreateFromBase64FileBuffer() static `OCRStudioSDKImage*` `ocrstudio::OCRStudioSDKImage::CreateFromBase64FileBuffer` (
 const char * *base64_data*,
 int *page_number* = 0,
 int *max_width* = 25000,
 int *max_height* = 25000) [static]

Creates an image from file loaded in a buffer encoded in base64.

Parameters

<i>base64_data</i>	- file buffer encoded as a base64 string
<i>page_number</i>	- page number, starting with 0
<i>max_width</i>	- maximum image width in pixels (0 for unrestricted)
<i>max_height</i>	- maximum image height in pixels (0 for unrestricted)

Returns

Pointer to a new image, the ownership is relinquished.

1.3.2.7 CreateFromPixelBuffer() static `OCRStudioSDKImage*` `ocrstudio::OCRStudioSDKImage::CreateFromPixelBuffer` (
 unsigned char * *data*,
 int *data_size*,
 int *width*,
 int *height*,
 int *bytes_per_line*,

```
int bytes_per_channel,  
OCRStudioSDKPixelFormat pixel_format ) [static]
```

Creates an image from a pixel buffer, the content is copied.

Parameters

<i>data</i>	- pointer to a pixels buffer
<i>data_size</i>	- size of the pixels buffer
<i>width</i>	- width of the image in pixels
<i>height</i>	- height of the image in pixels
<i>bytes_per_line</i>	- size of an image row in bytes (including alignment)
<i>bytes_per_channel</i>	- size of a pixel component in bytes
<i>pixel_format</i>	- pixel format

Returns

Pointer to a new image, the ownership is relinquished.

1.3.2.8 CreateFromBuffer() static `OCRStudioSDKImage*` `ocrstudio::OCRStudioSDKImage::CreateFromBuffer (`
`unsigned char * data,`
`int data_size,`
`int width,`
`int height,`
`int bytes_per_line,`
`int channels ) [static]`

Creates an image from a buffer, the content is copied.

Parameters

<i>data</i>	- pointer to a pixels buffer
<i>data_size</i>	- size of the pixels buffer
<i>width</i>	- width of the image in pixels
<i>height</i>	- height of the image in pixels
<i>bytes_per_line</i>	- size of an image row in bytes (including alignment)
<i>channels</i>	number of channels per-pixel

Returns

Pointer to a new image, the ownership is relinquished.

1.3.2.9 CreateFromYUVSimple() static `OCRStudioSDKImage*` `ocrstudio::OCRStudioSDKImage::CreateFromYUVSimple (`
`unsigned char * yuv_data,`
`int yuv_data_size,`
`int width,`
`int height ) [static]`

Creates an image from a simple YUV NV21 buffer.

Parameters

<i>yuv_data</i>	- pointer to YUV NV21 buffer
<i>yuv_data_size</i>	- size of the YUV NV21 buffer
<i>width</i>	- width of the image in pixels
<i>height</i>	- height of the image in pixels

Returns

Pointer to a new image, the ownership is relinquished.

1.3.2.10 CreateFromYUV() static `OCRStudioSDKImage*` `ocrstudio::OCRStudioSDKImage::CreateFromYUV`
(

```
    unsigned char * y_plane,
    int y_plane_size,
    int y_plane_row_stride,
    int y_plane_pixel_stride,
    unsigned char * u_plane,
    int u_plane_size,
    int u_plane_row_stride,
    int u_plane_pixel_stride,
    unsigned char * v_plane,
    int v_plane_size,
    int v_plane_row_stride,
    int v_plane_pixel_stride,
    int width,
    int height,
    OCRStudioSDKYUVFormat yuv_format ) [static]
```

Creates an image from a universal YUV buffer.

Parameters

<i>y_plane</i>	- pointer to Y plane buffer
<i>y_plane_size</i>	- Y plane buffer size
<i>y_plane_row_stride</i>	- Y plane row stride
<i>y_plane_pixel_stride</i>	- Y plane pixel stride
<i>u_plane</i>	- pointer to U plane buffer
<i>u_plane_size</i>	- U plane buffer size
<i>u_plane_row_stride</i>	- U plane row stride
<i>u_plane_pixel_stride</i>	- U plane pixel stride
<i>v_plane</i>	- pointer to V plane buffer
<i>v_plane_size</i>	- V plane buffer size
<i>v_plane_row_stride</i>	- V plane row stride
<i>v_plane_pixel_stride</i>	- V plane pixel stride
<i>width</i>	- image width in pixels
<i>height</i>	- image height in pixels
<i>yuv_format</i>	- YUV format specification

Returns

Pointer to a new image, the ownership is relinquished.

1.3.2.11 DeepCopy() `virtual OCRStudioSDKImage* ocrstudio::OCRStudioSDKImage::DeepCopy ()
const [pure virtual]`

Copies an image with copying of all pixels.

Returns

Pointer to a new copied image, the ownership is relinquished.

1.3.2.12 ShallowCopy() `virtual OCRStudioSDKImage* ocrstudio::OCRStudioSDKImage::ShallowCopy ( ) const [pure virtual]`

Copies an image without copying the pixels, retaining internal memory reference. The operations with the copied image will be invalid after the source is deleted.

Returns

Pointer to a new copied image, the ownership is relinquished.

1.3.2.13 ExportPixelBufferLength() `virtual int ocrstudio::OCRStudioSDKImage::ExportPixelBuffer↵
Length () const [pure virtual]`

Returns the required size of the export pixel buffer.

Returns

Number of required bytes

1.3.2.14 ExportPixelBuffer() `virtual int ocrstudio::OCRStudioSDKImage::ExportPixelBuffer (
 unsigned char * export_buffer,
 int export_buffer_length) const [pure virtual]`

Copies the pixels into an external buffer. For any image the exported buffer pixels will have 8-bit channels (0 means lowest intensity, 255 means highest intensity). 1-channel images are exported as grayscale, 3-channel images are exported as RGB, other images are copied as-is.

Parameters

<i>export_buffer</i>	- pointer to an output pixels buffer
<i>export_buffer_length</i>	- available buffer size. Must be at least the size returned by the ExportPixelBufferLength() method.

Returns

The number of written bytes

1.3.2.15 ExportBase64JPEG() virtual [OCRStudioSDKString](#) ocrstudio::OCRStudioSDKImage::ExportBase64JPEG () const [pure virtual]

Exports image as a JPEG buffer encoded in base64.

Returns

Base64 JPEG encoding of an image in a [OCRStudioSDKString](#) form

1.3.2.16 Scale() virtual void ocrstudio::OCRStudioSDKImage::Scale (
 int *width*,
 int *height*) [pure virtual]

Scales the internal image to a new size.

Parameters

<i>width</i>	- new width of the image in pixels
<i>height</i>	- new height of the image in pixels

1.3.2.17 DeepCopyScaled() virtual [OCRStudioSDKImage*](#) ocrstudio::OCRStudioSDKImage::DeepCopyScaled (
 int *width*,
 int *height*) const [pure virtual]

Copies the image with scaling to a new size.

Parameters

<i>width</i>	- new width of the image in pixels
<i>height</i>	- new height of the image in pixels

Returns

Pointer to a new scaled image, the ownership is relinquished.

1.3.2.18 CropByQuad() `virtual void ocrstudio::OCRStudioSDKImage::CropByQuad (`
 `const char * quad_json,`
 `int width,`
 `int height ) [pure virtual]`

Crops an image quadrilateral to a new image, with a new provided size. If width or height is less or equal to zero, the size will be calculated approximately based on an input quadrilateral.

Parameters

<i>quad_json</i>	- JSON representation of a quadrangle coordinates, in form [[x1, y1], [x2, y2], [x3, y3], [x4, y4]]
<i>width</i>	- new width of the image in pixels (or <= 0 for size autoselection)
<i>height</i>	- new height of the image in pixels (or <= 0 for size autoselection)

1.3.2.19 DeepCopyCroppedByQuad() `virtual OCRStudioSDKImage* ocrstudio::OCRStudioSDKImage::↔`
`DeepCopyCroppedByQuad (`
 `const char * quad_json,`
 `int width,`
 `int height ) const [pure virtual]`

Copies an image cropped by a quadrilateral, with a new provided size. If width or height is less or equal to zero, the size will be calculated approximately based on an input quadrilateral.

Parameters

<i>quad_json</i>	- JSON representation of a quadrangle coordinates, in form [[x1, y1], [x2, y2], [x3, y3], [x4, y4]]
<i>width</i>	- new width of the image in pixels (or <= 0 for size autoselection)
<i>height</i>	- new height of the image in pixels (or <= 0 for size autoselection)

Returns

Pointer to a new cropped image, the ownership is relinquished.

1.3.2.20 CropByRect() `virtual void ocrstudio::OCRStudioSDKImage::CropByRect (`
 `int x,`
 `int y,`
 `int width,`
 `int height ) [pure virtual]`

Crops an image to a rectangular region.

Parameters

<i>x</i>	- horizontal coordinate of the top-left corner
<i>y</i>	- vertical coordinate of the top-left corner
<i>width</i>	- width of the rectangle
<i>height</i>	- height of the rectangle

1.3.2.21 DeepCopyCroppedByRect() virtual [OCRStudioSDKImage*](#) ocrstudio::OCRStudioSDKImage::↔

```

DeepCopyCroppedByRect (
    int x,
    int y,
    int width,
    int height ) const [pure virtual]

```

Copies an image cropped to a rectangular region.

Parameters

<i>x</i>	- horizontal coordinate of the top-left corner
<i>y</i>	- vertical coordinate of the top-left corner
<i>width</i>	- width of the rectangle
<i>height</i>	- height of the rectangle

Returns

Pointer to a new cropped image, the ownership is relinquished.

1.3.2.22 ShallowCopyCroppedByRect() virtual [OCRStudioSDKImage*](#) ocrstudio::OCRStudioSDKImage↔

```

::ShallowCopyCroppedByRect (
    int x,
    int y,
    int width,
    int height ) const [pure virtual]

```

Shallow-copies an image cropped to a rectangular region. Operations on the resulting image are invalid after the source image is deleted.

Parameters

<i>x</i>	- horizontal coordinate of the top-left corner
<i>y</i>	- vertical coordinate of the top-left corner
<i>width</i>	- width of the rectangle
<i>height</i>	- height of the rectangle

Returns

Pointer to a new cropped image, the ownership is relinquished.

1.3.2.23 RotateByNinety() `virtual void ocrstudio::OCRStudioSDKImage::RotateByNinety (int num_rotations ) [pure virtual]`

Rotates the image clockwise by 90 degrees.

Parameters

<code>num_rotations</code>	- the number of times the rotation is performed
----------------------------	---

1.3.2.24 DeepCopyRotatedByNinety() `virtual OCRStudioSDKImage* ocrstudio::OCRStudioSDKImage::↔DeepCopyRotatedByNinety (int num_rotations ) const [pure virtual]`

Copies the image rotated clockwise by 90 degrees.

Parameters

<code>num_rotations</code>	- the number of times the rotation is performed
----------------------------	---

Returns

Pointer to a new rotated image, the ownership is relinquished.

1.4 ocrstudio::OCRStudioSDKInstance Class Reference

Main recognition engine class containing configuration for creating recognition sessions.

Public Member Functions

- virtual `~OCRStudioSDKInstance` ()=default
Default destructor.
- virtual const char * `Description` () const =0
Returns a description of a configured engine in JSON format.
- virtual `OCRStudioSDKSession` * `CreateSession` (const char *authorization_signature, const char *json_↔session_params, `OCRStudioSDKDelegate` *callback_delegate=nullptr) const =0
Creates a processing session with the provided parameters.

Static Public Member Functions

- static [OCRStudioSDKInstance](#) * [CreateStandalone](#) (const char *json_instance_init_params=nullptr)
Creates a new recognition engine instance from an internal configuration, embedded inside the library, if one is available. If no configuration is embedded inside, the method will throw an exception.
- static [OCRStudioSDKInstance](#) * [CreateFromPath](#) (const char *configuration_filename, const char *json_instance_init_params=nullptr)
Creates a new recognition engine instance from a configuration file (a binary file with an extension '.ocr').
- static [OCRStudioSDKInstance](#) * [CreateFromBuffer](#) (unsigned char *configuration_buffer, int configuration_buffer_size, const char *json_instance_init_params=nullptr)
Creates a new recognition engine instance from a configuration buffer (a binary buffer where the configuration file is loaded).
- static const char * [LibraryVersion](#) ()
Returns a string representation of the OCRStudioSDK library version.

1.4.1 Detailed Description

Main recognition engine class containing configuration for creating recognition sessions.

Definition at line 25 of file [ocr_studio_instance.h](#).

1.4.2 Member Function Documentation

1.4.2.1 CreateStandalone() static [OCRStudioSDKInstance](#)* ocrstudio::OCRStudioSDKInstance::CreateStandalone (const char * json_instance_init_params = nullptr) [static]

Creates a new recognition engine instance from an internal configuration, embedded inside the library, if one is available. If no configuration is embedded inside, the method will throw an exception.

Parameters

<i>json_instance_init_params</i>	- optional JSON with initialization parameters, in the following format (all keys are optional): { "enable_lazy_initialization": (bool), "enable_delayed_initialization": (bool), "initialization_num_threads": (int >= 0) }
----------------------------------	--

Returns

Pointer to a new instance object, the ownership is relinquished.

1.4.2.2 CreateFromPath() static [OCRStudioSDKInstance](#)* ocrstudio::OCRStudioSDKInstance::CreateFromPath (const char * configuration_filename, const char * json_instance_init_params = nullptr) [static]

Creates a new recognition engine instance from a configuration file (a binary file with an extension '.ocr').

Parameters

<i>configuration_filename</i>	- path to a configuration file *.ocr
<i>json_instance_init_params</i>	- optional JSON with initialization parameters, in the following format (all keys are optional): { "enable_lazy_initialization": (bool), "enable_delayed_initialization": (bool), "initialization_num_threads": (int >= 0) }

Returns

Pointer to a new instance object, the ownership is relinquished.

1.4.2.3 CreateFromBuffer() `static OCRStudioSDKInstance* ocrstudio::OCRStudioSDKInstance::CreateFromBuffer (`
`unsigned char * configuration_buffer,`
`int configuration_buffer_size,`
`const char * json_instance_init_params = nullptr ) [static]`

Creates a new recognition engine instance from a configuration buffer (a binary buffer where the configuration file is loaded).

Parameters

<i>configuration_buffer</i>	- pointer to a binary configuration buffer
<i>configuration_buffer_size</i>	- size of the configuration buffer in bytes
<i>json_instance_init_params</i>	- optional JSON with initialization parameters, in the following format (all keys are optional): { "enable_lazy_initialization": (bool), "enable_delayed_initialization": (bool), "initialization_num_threads": (int >= 0) }

Returns

Pointer to a new instance object, the ownership is relinquished.

1.4.2.4 Description() `virtual const char* ocrstudio::OCRStudioSDKInstance::Description ( ) const`
`[pure virtual]`

Returns a description of a configured engine in JSON format.

Returns

a JSON description in the following format: { "session_types": [(list of available session types)], "target_groups": [// present if there are target-oriented sessions { "target_group_type": "(group_type_name)", "targets": ["(target_name)", ...], "target_masks": ["(target_mask)", ...] }, ...] }

1.4.2.5 CreateSession() virtual [OCRStudioSDKSession*](#) ocrstudio::OCRStudioSDKInstance::CreateSession (

```

    const char * authorization_signature,
    const char * json_session_params,
    OCRStudioSDKDelegate * callback_delegate = nullptr ) const [pure virtual]

```

Creates a processing session with the provided parameters.

Parameters

<i>authorization_signature</i>	- signature of an authorized SDK user
<i>json_session_params</i>	- parameters of the created session, encoded in JSON in the following format: { "session_type": "(session_type)", "target_group_type": "(group_type_name)", "target_masks": ["(target_name_or_mask", ...], // optional, single string permitted "options": { // optional "(option_name)": "(option_value)", ... }, "output_modes": ["(output_mode)", ...] } Possible variants for "output_mode" are "character_alternatives" and "field_geometry".
<i>callback_delegate</i>	- optional pointer to an implemented instance of a delegate for receiving runtime messages.

Returns

Pointer to a new session, the ownership is relinquished.

1.5 ocrstudio::OCRStudioSDKItem Class Reference

A constituent object of a recognized or analyzed target.

Public Member Functions

- virtual [~OCRStudioSDKItem](#) ()=default
Default destructor.
- virtual [OCRStudioSDKItem](#) * [DeepCopy](#) () const =0
Copies an item with copying of all internal information.
- virtual const char * [Type](#) () const =0
Returns the type of the item.
- virtual const char * [Name](#) () const =0
Returns the name of the item.
- virtual const char * [Value](#) () const =0
Returns the string representatio of the value of the item.
- virtual double [Confidence](#) () const =0
Returns the item confidence value (doubole in range [0.0, 1.0])
- virtual bool [Accepted](#) () const =0
Returns the item accept flag.
- virtual const char * [Attributes](#) () const =0
Returns the attributes of the item in JSON format.
- virtual bool [HasImage](#) () const =0
Returns true iff the item has an associated image.
- virtual const [OCRStudioSDKImage](#) & [Image](#) () const =0
Returns the associated image.
- virtual const char * [Description](#) () const =0
Returns a detailed JSON description (format depends on the type)

1.5.1 Detailed Description

A constituent object of a recognized or analyzed target.

Definition at line 24 of file [ocr_studio_result.h](#).

1.5.2 Member Function Documentation

1.5.2.1 DeepCopy() `virtual OCRStudioSDKItem* ocrstudio::OCRStudioSDKItem::DeepCopy ( ) const`
[pure virtual]

Copies an item with copying of all internal information.

Returns

Pointer to a new item structure, the ownership is relinquished.

1.5.2.2 Attributes() `virtual const char* ocrstudio::OCRStudioSDKItem::Attributes ( ) const` [pure virtual]

Returns the attributes of the item in JSON format.

Returns

a JSON attributes map in the following format: { "(attribute_name)": "(attribute_value)" }

1.6 ocrstudio::OCRStudioSDKItemIterator Class Reference

Map-like iterator for a collection of [OCRStudioSDKItem](#) objects.

Public Member Functions

- [~OCRStudioSDKItemIterator](#) ()
Non-trivial destructor.
- [OCRStudioSDKItemIterator](#) (const [OCRStudioSDKItemIterator](#) ©)
Copy constructor.
- [OCRStudioSDKItemIterator](#) & operator= (const [OCRStudioSDKItemIterator](#) &other)
Assignment operator.
- bool [IsEqualTo](#) (const [OCRStudioSDKItemIterator](#) &other) const
Returns true iff the instances point to the same item.
- bool [operator==](#) (const [OCRStudioSDKItemIterator](#) &other) const
Equality operator.
- bool [operator!=](#) (const [OCRStudioSDKItemIterator](#) &other) const
Inequality operator.
- [OCRStudioSDKItemIterator Next](#) () const
Returns the iterator to the next item in the collection.
- void [Step](#) ()
Moves the iterator to the next item in the collection.
- void [operator++](#) ()
Moves the iterator to the next item in the collection.
- const char * [Key](#) () const
Returns the key of the item in the collection.
- const [OCRStudioSDKItem](#) & [Item](#) () const
Returns the item to which the iterator points (const ref)

Static Public Member Functions

- static [OCRStudioSDKItemIterator](#) [CreateFromImplementation](#) (const [OCRStudioSDKItemIterator](#)↔
Implementation &rimpl)
Creates an [OCRStudioSDKItemIterator](#) object from its internal implementation.

Private Member Functions

- [OCRStudioSDKItemIterator](#) (const [OCRStudioSDKItemIteratorImplementation](#) &rimpl)
Private constructor from an internal implementation.

Private Attributes

- [OCRStudioSDKItemIteratorImplementation](#) * [pimpl_](#)
Internal implementation.

1.6.1 Detailed Description

Map-like iterator for a collection of [OCRStudioSDKItem](#) objects.

Definition at line 79 of file [ocr_studio_result.h](#).

1.7 ocrstudio::OCRStudioSDKResult Class Reference

Main session result class - container with full session result.

Public Member Functions

- virtual [~OCRStudioSDKResult](#) ()=default
Default destructor.
- virtual [OCRStudioSDKResult](#) * [DeepCopy](#) () const =0
Copies a result with copying of all internal information.
- virtual int [TargetsCount](#) () const =0
Returns the number of stored targets.
- virtual const [OCRStudioSDKTarget](#) & [TargetByIndex](#) (int target_index) const =0
Returns a specific stored target by its index.
- virtual bool [AllTargetsFinal](#) () const =0
Returns true if all targets can be considered final.
- virtual [OCRStudioSDKString](#) [Serialize](#) () const =0
Serialize to JSON object current result.

1.7.1 Detailed Description

Main session result class - container with full session result.

Definition at line 214 of file [ocr_studio_result.h](#).

1.7.2 Member Function Documentation

1.7.2.1 DeepCopy() `virtual OCRStudioSDKResult* ocrstudio::OCRStudioSDKResult::DeepCopy ( ) const [pure virtual]`

Copies a result with copying of all internal information.

Returns

Pointer to a new result structure, the ownership is relinquished.

1.7.2.2 TargetsCount() `virtual int ocrstudio::OCRStudioSDKResult::TargetsCount ( ) const [pure virtual]`

Returns the number of stored targets.

Returns

The number of stored targets

1.7.2.3 TargetByIndex() `virtual const OCRStudioSDKTarget& ocrstudio::OCRStudioSDKResult::↔
TargetByIndex (
int target_index) const [pure virtual]`

Returns a specific stored target by its index.

Parameters

<i>target_index</i>	- 0-based index of a stored target
---------------------	------------------------------------

Returns

Specific stored target (constant reference)

1.7.2.4 AllTargetsFinal() `virtual bool ocrstudio::OCRStudioSDKResult::AllTargetsFinal ( ) const [pure virtual]`

Returns true if all targets can be considered final.

Returns

All targets can be considered final

1.7.2.5 Serialize() virtual [OCRStudioSDKString](#) ocrstudio::OCRStudioSDKResult::Serialize ()
const [pure virtual]

Serialize to JSON object current result.

Returns

JSON string with current result

1.8 ocrstudio::OCRStudioSDKSession Class Reference

Main processing session class - agent for performing image analysis.

Public Member Functions

- virtual [~OCRStudioSDKSession](#) ()=default
Default destructor.
- virtual const char * [Description](#) () const =0
Returns a description of a created session in JSON format.
- virtual void [ProcessImage](#) (const [OCRStudioSDKImage](#) &image)=0
Processes an input image or video frame, updates the internal session state.
- virtual void [ProcessData](#) (const char *data_str)=0
Processes an input json as a string.
- virtual const [OCRStudioSDKResult](#) & [CurrentResult](#) () const =0
Returns the current accumulated result.
- virtual void [Reset](#) ()=0
Resets the state of the session to the initial one.
- virtual void [Suspend](#) ()=0
Suspend the session.
- virtual void [Resume](#) ()=0
Resume the session.

1.8.1 Detailed Description

Main processing session class - agent for performing image analysis.

Definition at line 24 of file [ocr_studio_session.h](#).

1.8.2 Member Function Documentation

1.8.2.1 Description() `virtual const char* ocrstudio::OCRStudioSDKSession::Description ( ) const`
[pure virtual]

Returns a description of a created session in JSON format.

Returns

a JSON description in the following format: { "session_type": "(session_type)", "target_group_type": "(group_type_name)", "targets": ["(target_name)", ...], "options": { "(option_name)": "(option_value)", ... }, "output_modes": ["(output_mode)", ...] }

1.8.2.2 ProcessImage() `virtual void ocrstudio::OCRStudioSDKSession::ProcessImage (`
`const OCRStudioSDKImage & image )` [pure virtual]

Processes an input image or video frame, updates the internal session state.

Parameters

<i>image</i>	- the input image to be processed
--------------	-----------------------------------

1.8.2.3 ProcessData() `virtual void ocrstudio::OCRStudioSDKSession::ProcessData ( const char * data_str ) [pure virtual]`

Processes an input json as a string.

Parameters

<i>data_str</i>	- the input JSON containing a description of mrz and photo in the following format: { "doc_type": "(doc_type)", "physical_fields": { "rfid_mrz": { "value": "(mrz)", "type": "String" }, "rfid_photo": { "value": "(photo_string)", "type": "Image" } } }
-----------------	---

1.8.2.4 CurrentResult() `virtual const OCRStudioSDKResult& ocrstudio::OCRStudioSDKSession::↔ CurrentResult ( ) const [pure virtual]`

Returns the current accumulated result.

Returns

Current accumulated session result (constant reference to an internal structure, the memory is owned by the session)

1.9 ocrstudio::OCRStudioSDKString Class Reference

Public Member Functions

- [~OCRStudioSDKString \(\)](#)
Non-trivial destructor.
- [OCRStudioSDKString \(\)](#)
Default constructor.
- [OCRStudioSDKString \(const char *c_str\)](#)
Constructor from a C-string.
- [OCRStudioSDKString \(const OCRStudioSDKString ©\)](#)
Copy constructor.
- [OCRStudioSDKString & operator= \(const OCRStudioSDKString &other\)](#)
Assignment operator.
- [OCRStudioSDKString & operator+= \(const OCRStudioSDKString &other\)](#)
Inplace concatenation.
- [OCRStudioSDKString operator+ \(const OCRStudioSDKString &other\) const](#)
General concatenation.
- `const char * CStr \(\) const`
Returns internal c-string.
- `int Size \(\) const`
Returns number of bytes stored.

Private Attributes

- int [size_](#)
length of the internal string in bytes
- char * [str_](#)
internal c-string

1.9.1 Detailed Description

Definition at line 19 of file [ocr_studio_string.h](#).

1.10 ocrstudio::OCRStudioSDKTarget Class Reference

Recognition or analysis target (document or other object)

Public Member Functions

- virtual [~OCRStudioSDKTarget](#) ()=default
Default destructor.
- virtual [OCRStudioSDKTarget](#) * [DeepCopy](#) () const =0
Copies a target with copying of all internal information.
- virtual const char * [Description](#) () const =0
Returns a description of a target in JSON format.
- virtual int [ItemsCountByType](#) (const char *item_type) const =0
Returns the number of items with a provided item type.
- virtual bool [HasItem](#) (const char *item_type, const char *item_name) const =0
Checks whether there is an item of a specified type with a specified item name.
- virtual const [OCRStudioSDKItem](#) & [Item](#) (const char *item_type, const char *item_name) const =0
Returns a specific item.
- virtual [OCRStudioSDKItemIterator](#) [ItemsBegin](#) (const char *item_type) const =0
Returns a map-like iterator to the start of the collection of items with the specified type.
- virtual [OCRStudioSDKItemIterator](#) [ItemsEnd](#) (const char *item_type) const =0
Returns a map-like iterator to the end of the collection of items with the specified type.
- virtual bool [IsFinal](#) () const =0
Returns true if the target can be considered final.

1.10.1 Detailed Description

Recognition or analysis target (document or other object)

Definition at line 132 of file [ocr_studio_result.h](#).

1.10.2 Member Function Documentation

1.10.2.1 DeepCopy() virtual `OCRStudioSDKTarget*` ocrstudio::OCRStudioSDKTarget::DeepCopy ()
const [pure virtual]

Copies a target with copying of all internal information.

Returns

Pointer to a new target structure, the ownership is relinquished.

1.10.2.2 Description() virtual const char* ocrstudio::OCRStudioSDKTarget::Description () const
[pure virtual]

Returns a description of a target in JSON format.

Returns

a JSON description in the following format: { "target_type": "(target_type_name)", "specific_type": "(specific_type_name)", "item_types": ["(item_type_name)", ...], "attributes": { "(attribute_name)": "(attribute_value)", ... } }

1.10.2.3 ItemsCountByType() virtual int ocrstudio::OCRStudioSDKTarget::ItemsCountByType (const char * item_type) const [pure virtual]

Returns the number of items with a provided item type.

Parameters

<i>item_type</i>	- name of the item type
------------------	-------------------------

Returns

The number of items of the specified type. The number of items is zero if the stored collection has zero size or if the specified item type is not supported for the returned target type

1.10.2.4 HasItem() virtual bool ocrstudio::OCRStudioSDKTarget::HasItem (const char * item_type, const char * item_name) const [pure virtual]

Checks whether there is an item of a specified type with a specified item name.

Parameters

<i>item_type</i>	- name of the item type
<i>item_name</i>	- name of the specific item

Returns

true iff there exists an item with a provided name in the collection of items of the provided type

1.10.2.5 Item() `virtual const OCRStudioSDKItem& ocrstudio::OCRStudioSDKTarget::Item (`
 `const char * item_type,`
 `const char * item_name ) const [pure virtual]`

Returns a specific item.

Parameters

<i>item_type</i>	- name of the item type
<i>item_name</i>	- name of the specific item

Returns

Specific item object (constant reference)

1.10.2.6 ItemsBegin() `virtual OCRStudioSDKItemIterator ocrstudio::OCRStudioSDKTarget::Items←`
`Begin (`
 `const char * item_type ) const [pure virtual]`

Returns a map-like iterator to the start of the collection of items with the specified type.

Parameters

<i>item_type</i>	- name of the item type
------------------	-------------------------

Returns

A map-like 'begin' iterator to the collection of items

1.10.2.7 ItemsEnd() `virtual OCRStudioSDKItemIterator ocrstudio::OCRStudioSDKTarget::ItemsEnd (`
 `const char * item_type ) const [pure virtual]`

Returns a map-like iterator to the end of the collection of items with the specified type.

Parameters

<i>item_type</i>	- name of the item type
------------------	-------------------------

Returns

A map-like 'end' iterator to the collection of items

1.10.2.8 IsFinal() `virtual bool ocrstudio::OCRStudioSDKTarget::IsFinal ( ) const [pure virtual]`

Returns true if the target can be considered final.

Returns

Can the target be considered final

2 File Documentation

2.1 ocr_studio_delegate.h File Reference

Feedback base class, allows to receive runtime messages from OCRStudioSDKSession.

Classes

- class [ocrstudio::OCRStudioSDKDelegate](#)

2.1.1 Detailed Description

Feedback base class, allows to receive runtime messages from OCRStudioSDKSession.

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Definition in file [ocr_studio_delegate.h](#).

2.2 ocr_studio_delegate.h

```
00001
00011 #pragma once
00012 #ifndef OCRSTUDIOSDK_OCR_STUDIO_DELEGATE_H_INCLUDED
00013 #define OCRSTUDIOSDK_OCR_STUDIO_DELEGATE_H_INCLUDED
00014
00015 #include <ocrstudiosdk/ocr_studio_export.h>
00016
00017 namespace ocrstudio {
00018
00019 class OCR_STUDIO_SDK_DLL_EXPORT OCRStudioSDKDelegate {
00020 public:
00022     virtual ~OCRStudioSDKDelegate() = default;
00023
00028     virtual void Callback(const char* json_message) = 0;
00029 };
00030
00031 } // namespace ocrstudio
00032
00033 #endif // OCRSTUDIOSDK_OCR_STUDIO_DELEGATE_H_INCLUDED
```

2.3 ocr_studio_exception.h File Reference

Main C++ exception class.

Classes

- class [ocrstudio::OCRStudioSDKException](#)

2.3.1 Detailed Description

Main C++ exception class.

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Definition in file [ocr_studio_exception.h](#).

2.4 ocr_studio_exception.h

```

00001
00011 #pragma once
00012 #ifndef OCRSTUDIOSDK_OCR_STUDIO_EXCEPTION_H_INCLUDED
00013 #define OCRSTUDIOSDK_OCR_STUDIO_EXCEPTION_H_INCLUDED
00014
00015 #include <ocrstudiosdk/ocr_studio_export.h>
00016
00017 namespace ocrstudio {
00018
00019 class OCR_STUDIO_SDK_DLL_EXPORT OCRStudioSDKException {
00020 public:
00022     virtual ~OCRStudioSDKException();
00023
00025     OCRStudioSDKException(
00026         const char* type, const char* msg);
00027
00029     OCRStudioSDKException(
00030         const OCRStudioSDKException& copy);
00031
00033     const char* Type() const;
00034
00036     const char* Message() const;
00037
00038 private:
00039     char* type_;
00040     char* msg_;
00041 };
00042
00043 } // namespace ocrstudio
00044
00045 #endif // OCRSTUDIOSDK_OCR_STUDIO_EXCEPTION_H_INCLUDED

```

2.5 ocr_studio_export.h File Reference

Common definitions for library exports.

2.5.1 Detailed Description

Common definitions for library exports.

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Definition in file [ocr_studio_export.h](#).

2.6 ocr_studio_export.h

```

00001
00011 #pragma once
00012 #ifndef OCRSTUDIOSDK_OCR_STUDIO_EXPORT_H_INCLUDED
00013 #define OCRSTUDIOSDK_OCR_STUDIO_EXPORT_H_INCLUDED
00014
00015 #if defined _WIN32
00016 # define OCR_STUDIO_SDK_DLL_EXPORT __declspec(dllexport)
00017 #else // defined _WIN32
00018 # if defined(__clang__) || defined(__GNUC__)
00019 # define OCR_STUDIO_SDK_DLL_EXPORT __attribute__((visibility ("default")))
00020 # else // clang of gnu
00021 # define OCR_STUDIO_SDK_DLL_EXPORT
00022 # endif // clang of gnu
00023 #endif // defined _WIN32
00024
00025 #endif // OCRSTUDIOSDK_OCR_STUDIO_EXPORT_H_INCLUDED

```

2.7 ocr_studio_image.h File Reference

Common image manipulation facilities.

Classes

- class [ocrstudio::OCRStudioSDKImage](#)
Bitmap image class.

Variables

- [OCRSTUDIOSDK_PIXEL_FORMAT_G](#) = 0
Greyscale.
- [OCRSTUDIOSDK_PIXEL_FORMAT_GA](#)
Greyscale + Alpha.
- [OCRSTUDIOSDK_PIXEL_FORMAT_AG](#)
Alpha + Greyscale.
- [OCRSTUDIOSDK_PIXEL_FORMAT_RGB](#)
RGB.
- [OCRSTUDIOSDK_PIXEL_FORMAT_BGR](#)
BGR.
- [OCRSTUDIOSDK_PIXEL_FORMAT_BGRA](#)
BGR + Alpha.
- [OCRSTUDIOSDK_PIXEL_FORMAT_ARGB](#)
Alpha + RGB.
- [OCRSTUDIOSDK_YUV_FORMAT_NOT_SET](#) = 0
Not set.
- [OCRSTUDIOSDK_YUV_FORMAT_NV21](#)
NV 21.

2.7.1 Detailed Description

Common image manipulation facilities.

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Definition in file [ocr_studio_image.h](#).

2.8 ocr_studio_image.h

```

00001
00011 #pragma once
00012 #ifndef OCRSTUDIOSDK_OCR_STUDIO_IMAGE_H_INCLUDED
00013 #define OCRSTUDIOSDK_OCR_STUDIO_IMAGE_H_INCLUDED
00014
00015 #include <ocrstudiosdk/ocr_studio_export.h>
00016 #include <ocrstudiosdk/ocr_studio_string.h>
00017
00018 namespace ocrstudio {
00019
00023 enum OCR_STUDIO_SDK_DLL_EXPORT OCRStudioSDKPixelFormat {
00024     OCRSTUDIOSDK_PIXEL_FORMAT_G = 0,
00025     OCRSTUDIOSDK_PIXEL_FORMAT_GA,
00026     OCRSTUDIOSDK_PIXEL_FORMAT_AG,
00027     OCRSTUDIOSDK_PIXEL_FORMAT_RGB,
00028     OCRSTUDIOSDK_PIXEL_FORMAT_BGR,
00029     OCRSTUDIOSDK_PIXEL_FORMAT_BGRA,
00030     OCRSTUDIOSDK_PIXEL_FORMAT_ARGB,
00031     OCRSTUDIOSDK_PIXEL_FORMAT_RGBA
00032 };
00033
00034
00035
00039 enum OCR_STUDIO_SDK_DLL_EXPORT OCRStudioSDKYUVFormat {
00040     OCRSTUDIOSDK_YUV_FORMAT_NOT_SET = 0,
00041     OCRSTUDIOSDK_YUV_FORMAT_NV21,
00042     OCRSTUDIOSDK_YUV_FORMAT_420_888
00043 };
00044
00045
00046
00050 class OCR_STUDIO_SDK_DLL_EXPORT OCRStudioSDKImage {
00051 public:
00057     static int PagesCount(const char* filename);
00058
00065     static OCRStudioSDKString PageName(
00066         const char *filename, int page_number);
00067
00068 public:
00073     static OCRStudioSDKImage* CreateEmpty();
00074
00083     static OCRStudioSDKImage* CreateFromFile(
00084         const char* filename,
00085         int page_number = 0,
00086         int max_width = 25000,
00087         int max_height = 25000);
00088
00098     static OCRStudioSDKImage* CreateFromFileBuffer(
00099         unsigned char* data,
00100         int data_size,
00101         int page_number = 0,
00102         int max_width = 25000,
00103         int max_height = 25000);
00104
00113     static OCRStudioSDKImage* CreateFromBase64FileBuffer(
00114         const char* base64_data,
00115         int page_number = 0,
00116         int max_width = 25000,
00117         int max_height = 25000);
00118
00130     static OCRStudioSDKImage* CreateFromPixelBuffer(
00131         unsigned char* data,
00132         int data_size,
00133         int width,
00134         int height,
00135         int bytes_per_line,
00136         int bytes_per_channel,
00137         OCRStudioSDKPixelFormat pixel_format);
00138
00149     static OCRStudioSDKImage* CreateFromBuffer(
00150         unsigned char* data,
00151         int data_size,
00152         int width,
00153         int height,
00154         int bytes_per_line,
00155         int channels);
00156
00165     static OCRStudioSDKImage* CreateFromYUVSimple(
00166         unsigned char* yuv_data,
00167         int yuv_data_size,
00168         int width,
00169         int height);
00170
00190     static OCRStudioSDKImage* CreateFromYUV(
00191         unsigned char* y_plane,

```

```

00192     int            y_plane_size,
00193     int            y_plane_row_stride,
00194     int            y_plane_pixel_stride,
00195     unsigned char* u_plane,
00196     int            u_plane_size,
00197     int            u_plane_row_stride,
00198     int            u_plane_pixel_stride,
00199     unsigned char* v_plane,
00200     int            v_plane_size,
00201     int            v_plane_row_stride,
00202     int            v_plane_pixel_stride,
00203     int            width,
00204     int            height,
00205     OCRStudioSDKYUVFormat yuv_format);
00206
00207 public:
00209     virtual ~OCRStudioSDKImage() = default;
00210
00215     virtual OCRStudioSDKImage* DeepCopy() const = 0;
00216
00223     virtual OCRStudioSDKImage* ShallowCopy() const = 0;
00224
00228     virtual void Clear() = 0;
00229
00234     virtual int ExportPixelBufferLength() const = 0;
00235
00246     virtual int ExportPixelBuffer(unsigned char* export_buffer, int export_buffer_length) const = 0;
00247
00252     virtual OCRStudioSDKString ExportBase64JPEG() const = 0;
00253
00259     virtual void Scale(int width, int height) = 0;
00260
00267     virtual OCRStudioSDKImage* DeepCopyScaled(int width, int height) const = 0;
00268
00279     virtual void CropByQuad(const char* quad_json, int width, int height) = 0;
00280
00291     virtual OCRStudioSDKImage* DeepCopyCroppedByQuad(
00292         const char* quad_json, int width, int height) const = 0;
00293
00301     virtual void CropByRect(int x, int y, int width, int height) = 0;
00302
00311     virtual OCRStudioSDKImage* DeepCopyCroppedByRect(
00312         int x, int y, int width, int height) const = 0;
00313
00324     virtual OCRStudioSDKImage* ShallowCopyCroppedByRect(
00325         int x, int y, int width, int height) const = 0;
00326
00331     virtual void RotateByNinety(int num_rotations) = 0;
00332
00338     virtual OCRStudioSDKImage* DeepCopyRotatedByNinety(int num_rotations) const = 0;
00339
00341     virtual int Width() const = 0;
00342
00344     virtual int Height() const = 0;
00345
00347     virtual int BytesPerLine() const = 0;
00348
00350     virtual int Channels() const = 0;
00351
00353     virtual void* UnsafeBufferPtr() const = 0;
00354
00356     virtual bool OwnsPixelData() const = 0;
00357
00359     virtual void ForcePixelDataOwnership() = 0;
00360 };
00361
00362 } // namespace ocrstudio
00364
00365 #endif // OCRSTUDIOSDK_OCR_STUDIO_IMAGE_H_INCLUDED

```

2.9 ocr_studio_instance.h File Reference

Main recognition engine instance class declaration.

Classes

- class [ocrstudio::OCRStudioSDKInstance](#)

Main recognition engine class containing configuration for creating recognition sessions.

2.9.1 Detailed Description

Main recognition engine instance class declaration.

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Definition in file [ocr_studio_instance.h](#).

2.10 ocr_studio_instance.h

```

00001
00011 #pragma once
00012 #ifndef OCRSTUDIOSDK_OCR_STUDIO_INSTANCE_H_INCLUDED
00013 #define OCRSTUDIOSDK_OCR_STUDIO_INSTANCE_H_INCLUDED
00014
00015 #include <ocrstudiosdk/ocr_studio_export.h>
00016 #include <ocrstudiosdk/ocr_studio_session.h>
00017 #include <ocrstudiosdk/ocr_studio_delegate.h>
00018
00019 namespace ocrstudio {
00020
00025 class OCR_STUDIO_SDK_DLL_EXPORT OCRStudioSDKInstance {
00026 public:
00040     static OCRStudioSDKInstance* CreateStandalone(
00041         const char* json_instance_init_params = nullptr);
00042
00056     static OCRStudioSDKInstance* CreateFromPath(
00057         const char* configuration_filename,
00058         const char* json_instance_init_params = nullptr);
00059
00074     static OCRStudioSDKInstance* CreateFromBuffer(
00075         unsigned char* configuration_buffer,
00076         int configuration_buffer_size,
00077         const char* json_instance_init_params = nullptr);
00078
00079 public:
00080
00084     static const char* LibraryVersion();
00085
00086 public:
00088     virtual ~OCRStudioSDKInstance() = default;
00089
00107     virtual const char* Description() const = 0;
00108
00129     virtual OCRStudioSDKSession* CreateSession(
00130         const char* authorization_signature,
00131         const char* json_session_params,
00132         OCRStudioSDKDelegate* callback_delegate = nullptr) const = 0;
00133 };
00134
00135 } // namespace ocrstudio
00136
00137 #endif // OCRSTUDIOSDK_OCR_STUDIO_INSTANCE_H_INCLUDED

```

2.11 ocr_studio_result.h File Reference

Result containers.

Classes

- class [ocrstudio::OCRStudioSDKItem](#)
A constituent object of a recognized or analyzed target.
- class [ocrstudio::OCRStudioSDKItemIterator](#)
Map-like iterator for a collection of [OCRStudioSDKItem](#) objects.
- class [ocrstudio::OCRStudioSDKTarget](#)
Recognition or analysis target (document or other object)
- class [ocrstudio::OCRStudioSDKResult](#)
Main session result class - container with full session result.

2.11.1 Detailed Description

Result containers.

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Definition in file [ocr_studio_result.h](#).

2.12 ocr_studio_result.h

```

00001
00011 #pragma once
00012 #ifndef OCRSTUDIOSDK_OCR_STUDIO_RESULT_H_INCLUDED
00013 #define OCRSTUDIOSDK_OCR_STUDIO_RESULT_H_INCLUDED
00014
00015 #include <ocrstudiosdk/ocr_studio_export.h>
00016 #include <ocrstudiosdk/ocr_studio_string.h>
00017 #include <ocrstudiosdk/ocr_studio_image.h>
00018
00019 namespace ocrstudio {
00020
00024 class OCR_STUDIO_SDK_DLL_EXPORT OCRStudioSDKItem {
00025 public:
00027     virtual ~OCRStudioSDKItem() = default;
00028
00033     virtual OCRStudioSDKItem* DeepCopy() const = 0;
00034
00036     virtual const char* Type() const = 0;
00037
00039     virtual const char* Name() const = 0;
00040
00042     virtual const char* Value() const = 0;
00043
00045     virtual double Confidence() const = 0;
00046
00048     virtual bool Accepted() const = 0;
00049
00057     virtual const char* Attributes() const = 0;
00058
00060     virtual bool HasImage() const = 0;
00061
00063     virtual const OCRStudioSDKImage& Image() const = 0;
00064
00066     virtual const char* Description() const = 0;
00067 };
00068
00069
00070
00074 class OCRStudioSDKItemIteratorImplementation;
00075
00079 class OCR_STUDIO_SDK_DLL_EXPORT OCRStudioSDKItemIterator {
00080 public:
00082     ~OCRStudioSDKItemIterator();
00083
00085     OCRStudioSDKItemIterator(const OCRStudioSDKItemIterator& copy);
00086
00088     OCRStudioSDKItemIterator& operator =(const OCRStudioSDKItemIterator& other);
00089
00091     bool IsEqualTo(const OCRStudioSDKItemIterator& other) const;
00092
00094     bool operator ==(const OCRStudioSDKItemIterator& other) const;
00095
00097     bool operator !=(const OCRStudioSDKItemIterator& other) const;
00098
00100     OCRStudioSDKItemIterator Next() const;
00101
00103     void Step();
00104
00106     void operator ++();
00107
00109     const char* Key() const;
00110
00112     const OCRStudioSDKItem& Item() const;
00113
00114 public:
00116     static OCRStudioSDKItemIterator CreateFromImplementation(
00117         const OCRStudioSDKItemIteratorImplementation& rimpl);
00118
00119 private:
00121     OCRStudioSDKItemIterator(const OCRStudioSDKItemIteratorImplementation& rimpl);

```

```

00122
00124     OCRStudioSDKItemIteratorImplementation* pimpl_;
00125 };
00126
00127
00128
00132 class OCR_STUDIO_SDK_DLL_EXPORT OCRStudioSDKTarget {
00133 public:
00135     virtual ~OCRStudioSDKTarget() = default;
00136
00141     virtual OCRStudioSDKTarget* DeepCopy() const = 0;
00142
00156     virtual const char* Description() const = 0;
00157
00165     virtual int ItemsCountByType(const char* item_type) const = 0;
00166
00175     virtual bool HasItem(const char* item_type, const char* item_name) const = 0;
00176
00183     virtual const OCRStudioSDKItem& Item(
00184         const char* item_type, const char* item_name) const = 0;
00185
00192     virtual OCRStudioSDKItemIterator ItemsBegin(const char* item_type) const = 0;
00193
00200     virtual OCRStudioSDKItemIterator ItemsEnd(const char* item_type) const = 0;
00201
00206     virtual bool IsFinal() const = 0;
00207 };
00208
00209
00210
00214 class OCR_STUDIO_SDK_DLL_EXPORT OCRStudioSDKResult {
00215 public:
00217     virtual ~OCRStudioSDKResult() = default;
00218
00223     virtual OCRStudioSDKResult* DeepCopy() const = 0;
00224
00229     virtual int TargetsCount() const = 0;
00230
00236     virtual const OCRStudioSDKTarget& TargetByIndex(int target_index) const = 0;
00237
00242     virtual bool AllTargetsFinal() const = 0;
00243
00248     virtual OCRStudioSDKString Serialize() const = 0;
00249 };
00250
00251 } // namespace ocrstudio
00252
00253 #endif // OCRSTUDIOSDK_OCR_STUDIO_RESULT_H_INCLUDED

```

2.13 ocr_studio_session.h File Reference

Main processing session class declaration.

Classes

- class [ocrstudio::OCRStudioSDKSession](#)
Main processing session class - agent for performing image analysis.

2.13.1 Detailed Description

Main processing session class declaration.

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Definition in file [ocr_studio_session.h](#).

2.14 ocr_studio_session.h

```

00001
00011 #pragma once
00012 #ifndef OCRSTUDIOSDK_OCR_STUDIO_SESSION_H_INCLUDED
00013 #define OCRSTUDIOSDK_OCR_STUDIO_SESSION_H_INCLUDED
00014
00015 #include <ocrstudiosdk/ocr_studio_export.h>
00016 #include <ocrstudiosdk/ocr_studio_image.h>
00017 #include <ocrstudiosdk/ocr_studio_result.h>
00018
00019 namespace ocrstudio {
00020
00024 class OCR_STUDIO_SDK_DLL_EXPORT OCRStudioSDKSession {
00025 public:
00027     virtual ~OCRStudioSDKSession() = default;
00028
00029 public:
00044     virtual const char* Description() const = 0;
00045
00051     virtual void ProcessImage(const OCRStudioSDKImage& image) = 0;
00052
00070     virtual void ProcessData(const char* data_str) = 0;
00071
00077     virtual const OCRStudioSDKResult& CurrentResult() const = 0;
00078
00082     virtual void Reset() = 0;
00083
00087     virtual void Suspend() = 0;
00088
00092     virtual void Resume() = 0;
00093
00094 };
00095
00096 } // namespace ocrstudio
00097
00098 #endif // OCRSTUDIOSDK_OCR_STUDIO_SESSION_H_INCLUDED

```

2.15 ocr_studio_string.h File Reference

String manipulation facilities.

Classes

- class [ocrstudio::OCRStudioSDKString](#)

2.15.1 Detailed Description

String manipulation facilities.

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Definition in file [ocr_studio_string.h](#).

2.16 ocr_studio_string.h

```

00001
00011 #pragma once
00012 #ifndef OCRSTUDIOSDK_OCR_STUDIO_STRING_H_INCLUDED
00013 #define OCRSTUDIOSDK_OCR_STUDIO_STRING_H_INCLUDED
00014
00015 #include <ocrstudiosdk/ocr_studio_export.h>
00016
00017 namespace ocrstudio {
00018
00019 class OCR_STUDIO_SDK_DLL_EXPORT OCRStudioSDKString {
00020 public:
00022     ~OCRStudioSDKString();
00023

```

```
00025   OCRStudioSDKString();
00026
00028   explicit OCRStudioSDKString(const char* c_str);
00029
00031   OCRStudioSDKString(
00032       const OCRStudioSDKString& copy);
00033
00035   OCRStudioSDKString& operator =(
00036       const OCRStudioSDKString& other);
00037
00039   OCRStudioSDKString& operator +=(
00040       const OCRStudioSDKString& other);
00041
00043   OCRStudioSDKString operator +(
00044       const OCRStudioSDKString& other) const;
00045
00047   const char* CStr() const;
00048
00050   int Size() const;
00051
00052 private:
00053     int size_;
00054     char* str_;
00055 };
00056
00057 } // namespace ocrstudio
00058
00059 #endif // OCRSTUDIOSDK_OCR_STUDIO_STRING_H_INCLUDED
```

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